
Introduction

When you analyze algorithms, you often need to draw upon a body of mathematical tools. Some of these tools are as simple as high-school algebra, but others may be new to you. In Part I, we saw how to manipulate asymptotic notations and solve recurrences. This appendix comprises a compendium of several other concepts and methods used in analyzing algorithms. As noted in the introduction to Part I, you may have seen much of the material in this appendix before having read this book, although some of the specific notational conventions appearing here might differ from those you have seen elsewhere. Hence, you should treat this appendix as reference material. As in the rest of this book, however, we have included exercises and problems, in order for you to improve your skills in these areas.

Appendix A offers methods for evaluating and bounding summations, which occur frequently in the analysis of algorithms. Many of the formulas here appear in any calculus text, but you will find it convenient to have these methods compiled in one place.

Appendix B contains basic definitions and notations for sets, relations, functions, graphs, and trees. It also gives some basic properties of these mathematical objects.

Appendix C begins with elementary principles of counting: permutations, combinations, and the like. The remainder contains definitions and properties of basic probability. Most of the algorithms in this book require no probability for their analysis, and thus you can easily omit the latter sections of the chapter on a first reading, even without skimming them. Later, when you encounter a probabilistic analysis that you want to understand better, you will find Appendix C well organized for reference purposes.

Appendix D defines matrices, their operations, and some of their basic properties. You have probably seen most of this material already if you have taken a course in linear algebra. But you might find it helpful to have one place to look for notations and definitions.

A Summations

When an algorithm contains an iterative control construct such as a **while** or **for** loop, you can express its running time as the sum of the times spent on each execution of the body of the loop. For example, Section 2.2 argued that the i th iteration of insertion sort took time proportional to i in the worst case. Adding up the time spent on each iteration produced the summation (or series) $\sum_{i=2}^n i$. Evaluating this summation resulted in a bound of $\Theta(n^2)$ on the worst-case running time of the algorithm. This example illustrates why you should know how to manipulate and bound summations.

Section A.1 lists several basic formulas involving summations. Section A.2 offers useful techniques for bounding summations. The formulas in Section A.1 appear without proof, though proofs for some of them appear in Section A.2 to illustrate the methods of that section. You can find most of the other proofs in any calculus text.

A.1 Summation formulas and properties

Given a sequence $a_1, a_2, \dots, a_n$ of numbers, where n is a nonnegative integer, the finite sum $a_1 + a_2 + \dots + a_n$ can be expressed as $\sum_{k=1}^n a_k$. If $n = 0$, the value of the summation is defined to be 0. The value of a finite series is always well defined, and the order in which its terms are added does not matter.

Given an infinite sequence $a_1, a_2, \dots$ of numbers, we can write their infinite sum $a_1 + a_2 + \dots$ as $\sum_{k=1}^{\infty} a_k$, which means $\lim_{n \rightarrow \infty} \sum_{k=1}^n a_k$. If the limit does not exist, the series *diverges*, and otherwise, it *converges*. The terms of a convergent series cannot always be added in any order. You can, however, rearrange the terms of an *absolutely convergent series*, that is, a series $\sum_{k=1}^{\infty} a_k$ for which the series $\sum_{k=1}^{\infty} |a_k|$ also converges.

Linearity

For any real number c and any finite sequences $a_1, a_2, \dots, a_n$ and $b_1, b_2, \dots, b_n$,

$$\sum_{k=1}^n (ca_k + b_k) = c \sum_{k=1}^n a_k + \sum_{k=1}^n b_k .$$

The linearity property also applies to infinite convergent series.

The linearity property applies to summations incorporating asymptotic notation. For example,

$$\sum_{k=1}^n \Theta(f(k)) = \Theta \left(\sum_{k=1}^n f(k) \right) .$$

In this equation, the Θ -notation on the left-hand side applies to the variable k , but on the right-hand side, it applies to n . Such manipulations also apply to infinite convergent series.

Arithmetic series

The summation

$$\sum_{k=1}^n k = 1 + 2 + \dots + n ,$$

is an *arithmetic series* and has the value

$$\sum_{k=1}^n k = \frac{n(n+1)}{2} \tag{A.1}$$

$$= \Theta(n^2) . \tag{A.2}$$

A *general arithmetic series* includes an additive constant $a \geq 0$ and a constant coefficient $b > 0$ in each term, but has the same total asymptotically:

$$\sum_{k=1}^n (a + bk) = \Theta(n^2) . \tag{A.3}$$

Sums of squares and cubes

The following formulas apply to summations of squares and cubes:

$$\sum_{k=0}^n k^2 = \frac{n(n+1)(2n+1)}{6} , \tag{A.4}$$

$$\sum_{k=0}^n k^3 = \frac{n^2(n+1)^2}{4} . \tag{A.5}$$

Geometric series

For real $x \neq 1$, the summation

$$\sum_{k=0}^n x^k = 1 + x + x^2 + \cdots + x^n$$

is a *geometric series* and has the value

$$\sum_{k=0}^n x^k = \frac{x^{n+1} - 1}{x - 1} . \quad (\text{A.6})$$

The infinite decreasing geometric series occurs when the summation is infinite and $|x| < 1$:

$$\sum_{k=0}^{\infty} x^k = \frac{1}{1 - x} . \quad (\text{A.7})$$

If we assume that $0^0 = 1$, these formulas apply even when $x = 0$.

Harmonic series

For positive integers n , the n th *harmonic number* is

$$\begin{aligned} H_n &= 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \cdots + \frac{1}{n} \\ &= \sum_{k=1}^n \frac{1}{k} \end{aligned} \quad (\text{A.8})$$

$$= \ln n + O(1) . \quad (\text{A.9})$$

Inequalities (A.20) and (A.21) on page 1150 provide the stronger bounds

$$\ln(n + 1) \leq H_n \leq \ln n + 1 . \quad (\text{A.10})$$

Integrating and differentiating series

Integrating or differentiating the formulas above yields additional formulas. For example, differentiating both sides of the infinite geometric series (A.7) and multiplying by x gives

$$\sum_{k=0}^{\infty} kx^k = \frac{x}{(1 - x)^2} \quad (\text{A.11})$$

for $|x| < 1$.

Telescoping series

For any sequence $a_0, a_1, \dots, a_n$,

$$\sum_{k=1}^n (a_k - a_{k-1}) = a_n - a_0, \quad (\text{A.12})$$

since each of the terms $a_1, a_2, \dots, a_{n-1}$ is added in exactly once and subtracted out exactly once. We say that the sum *telescopes*. Similarly,

$$\sum_{k=0}^{n-1} (a_k - a_{k+1}) = a_0 - a_n.$$

As an example of a telescoping sum, consider the series

$$\sum_{k=1}^{n-1} \frac{1}{k(k+1)}.$$

Rewriting each term as

$$\frac{1}{k(k+1)} = \frac{1}{k} - \frac{1}{k+1},$$

gives

$$\begin{aligned} \sum_{k=1}^{n-1} \frac{1}{k(k+1)} &= \sum_{k=1}^{n-1} \left(\frac{1}{k} - \frac{1}{k+1} \right) \\ &= 1 - \frac{1}{n}. \end{aligned}$$

Reindexing summations

A series can sometimes be simplified by changing its index, often reversing the order of summation. Consider the series $\sum_{k=0}^n a_{n-k}$. Because the terms in this summation are $a_n, a_{n-1}, \dots, a_0$, we can reverse the order of indices by letting $j = n - k$ and rewrite this summation as

$$\sum_{k=0}^n a_{n-k} = \sum_{j=0}^n a_j. \quad (\text{A.13})$$

Generally, if the summation index appears in the body of the sum with a minus sign, it's worth thinking about reindexing.

As an example, consider the summation

$$\sum_{k=1}^n \frac{1}{n-k+1}.$$

The index k appears with a negative sign in $1/(n-k+1)$. And indeed, we can simplify this summation, this time setting $j = n-k+1$, yielding

$$\sum_{k=1}^n \frac{1}{n-k+1} = \sum_{j=1}^n \frac{1}{j}, \quad (\text{A.14})$$

which is just the harmonic series (A.8).

Products

The finite product $a_1 a_2 \cdots a_n$ can be expressed as

$$\prod_{k=1}^n a_k.$$

If $n = 0$, the value of the product is defined to be 1. You can convert a formula with a product to a formula with a summation by using the identity

$$\lg \left(\prod_{k=1}^n a_k \right) = \sum_{k=1}^n \lg a_k.$$

Exercises

A.I-1

Prove that $\sum_{k=1}^n O(f_k(i)) = O(\sum_{k=1}^n f_k(i))$ by using the linearity property of summations.

A.I-2

Find a simple formula for $\sum_{k=1}^n (2k-1)$.

A.I-3

Interpret the decimal number 111,111,111 in light of equation (A.6).

A.I-4

Evaluate the infinite series $1 - \frac{1}{2} + \frac{1}{4} - \frac{1}{8} + \frac{1}{16} - \cdots$.

A.I-5

Let $c \geq 0$ be a constant. Show that $\sum_{k=1}^n k^c = \Theta(n^{c+1})$.

A.I-6

Show that $\sum_{k=0}^{\infty} k^2 x^k = x(1+x)/(1-x)^3$ for $|x| < 1$.

A.1-7

Prove that $\sum_{k=1}^n \sqrt{k \lg k} = \Theta(n^{3/2} \lg^{1/2} n)$. (*Hint: Show the asymptotic upper and lower bounds separately.*)

★ **A.1-8**

Show that $\sum_{k=1}^n 1/(2k-1) = \ln(\sqrt{n}) + O(1)$ by manipulating the harmonic series.

★ **A.1-9**

Show that $\sum_{k=0}^{\infty} (k-1)/2^k = 0$.

★ **A.1-10**

Evaluate the sum $\sum_{k=1}^{\infty} (2k+1)x^{2k}$ for $|x| < 1$.

★ **A.1-11**

Evaluate the product $\prod_{k=2}^n (1 - 1/k^2)$.

A.2 Bounding summations

You can choose from several techniques to bound the summations that describe the running times of algorithms. Here are some of the most frequently used methods.

Mathematical induction

The most basic way to evaluate a series is to use mathematical induction. As an example, let's prove that the arithmetic series $\sum_{k=1}^n k$ evaluates to $n(n+1)/2$. For $n = 1$, we have that $n(n+1)/2 = 1 \cdot 2/2 = 1$, which equals $\sum_{k=1}^1 k$. With the inductive assumption that it holds for n , we prove that it holds for $n+1$. We have

$$\begin{aligned} \sum_{k=1}^{n+1} k &= \sum_{k=1}^n k + (n+1) \\ &= \frac{n(n+1)}{2} + (n+1) \\ &= \frac{n^2 + n + 2n + 2}{2} \\ &= \frac{(n+1)(n+2)}{2}. \end{aligned}$$

You don't always need to guess the exact value of a summation in order to use mathematical induction. Instead, you can use induction to prove an upper or lower

bound on a summation. As an example, let's prove the asymptotic upper bound $\sum_{k=0}^n 3^k = O(3^n)$. More specifically, we'll prove that $\sum_{k=0}^n 3^k \leq c 3^n$ for some constant c . For the initial condition $n = 0$, we have $\sum_{k=0}^0 3^k = 1 \leq c \cdot 1$ as long as $c \geq 1$. Assuming that the bound holds for n , we prove that it holds for $n + 1$. We have

$$\begin{aligned} \sum_{k=0}^{n+1} 3^k &= \sum_{k=0}^n 3^k + 3^{n+1} \\ &\leq c 3^n + 3^{n+1} && \text{(by the inductive hypothesis)} \\ &= \left(\frac{1}{3} + \frac{1}{c}\right) c 3^{n+1} \\ &\leq c 3^{n+1} \end{aligned}$$

as long as $(1/3 + 1/c) \leq 1$ or, equivalently, $c \geq 3/2$. Thus, $\sum_{k=0}^n 3^k = O(3^n)$, as we wished to show.

You need to take care when using asymptotic notation to prove bounds by induction. Consider the following fallacious proof that $\sum_{k=1}^n k = O(n)$. Certainly, $\sum_{k=1}^1 k = O(1)$. Assuming that the bound holds for n , we now prove it for $n + 1$:

$$\begin{aligned} \sum_{k=1}^{n+1} k &= \sum_{k=1}^n k + (n + 1) \\ &= O(n) + (n + 1) && \Leftarrow \text{wrong!} \\ &= O(n + 1). \end{aligned}$$

The bug in the argument is that the “constant” hidden by the “big-oh” grows with n and thus is not constant. We have not shown that the same constant works for *all* n .

Bounding the terms

You can sometimes obtain a good upper bound on a series by bounding each term of the series, and it often suffices to use the largest term to bound the others. For example, a quick upper bound on the arithmetic series (A.1) is

$$\begin{aligned} \sum_{k=1}^n k &\leq \sum_{k=1}^n n \\ &= n^2. \end{aligned}$$

In general, for a series $\sum_{k=1}^n a_k$, if we let $a_{\max} = \max \{a_k : 1 \leq k \leq n\}$, then

$$\sum_{k=1}^n a_k \leq n \cdot a_{\max}.$$

The technique of bounding each term in a series by the largest term is a weak method when the series can in fact be bounded by a geometric series. Given the series $\sum_{k=0}^n a_k$, suppose that $a_{k+1}/a_k \leq r$ for all $k \geq 0$, where $0 < r < 1$ is a constant. You can bound the sum by an infinite decreasing geometric series, since $a_k \leq a_0 r^k$, and thus

$$\begin{aligned} \sum_{k=0}^n a_k &\leq \sum_{k=0}^{\infty} a_0 r^k \\ &= a_0 \sum_{k=0}^{\infty} r^k \end{aligned} \tag{A.15}$$

$$= a_0 \frac{1}{1-r} . \tag{A.16}$$

You can apply this method to bound the summation $\sum_{k=1}^{\infty} (k/3^k)$. In order to start the summation at $k = 0$, rewrite it as $\sum_{k=0}^{\infty} ((k+1)/3^{k+1})$. The first term (a_0) is $1/3$, and the ratio (r) of consecutive terms is

$$\begin{aligned} \frac{(k+2)/3^{k+2}}{(k+1)/3^{k+1}} &= \frac{1}{3} \cdot \frac{k+2}{k+1} \\ &\leq \frac{2}{3} \end{aligned}$$

for all $k \geq 0$. Thus, we have

$$\begin{aligned} \sum_{k=1}^{\infty} \frac{k}{3^k} &= \sum_{k=0}^{\infty} \frac{k+1}{3^{k+1}} \\ &\leq \frac{1}{3} \cdot \frac{1}{1-2/3} \\ &= 1 . \end{aligned}$$

A common bug in applying this method is to show that the ratio of consecutive terms is less than 1 and then to assume that the summation is bounded by a geometric series. An example is the infinite harmonic series, which diverges since

$$\begin{aligned} \sum_{k=1}^{\infty} \frac{1}{k} &= \lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{1}{k} \\ &= \lim_{n \rightarrow \infty} \Theta(\lg n) \\ &= \infty . \end{aligned}$$

The ratio of the $(k+1)$ st and k th terms in this series is $k/(k+1) < 1$, but the series is not bounded by a decreasing geometric series. To bound a series by a geometric

series, you need to show that there is an $r < 1$, which is a *constant*, such that the ratio of all pairs of consecutive terms never exceeds r . In the harmonic series, no such r exists because the ratio becomes arbitrarily close to 1.

Splitting summations

One way to obtain bounds on a difficult summation is to express the series as the sum of two or more series by partitioning the range of the index and then to bound each of the resulting series. For example, let's find a lower bound on the arithmetic series $\sum_{k=1}^n k$, which we have already seen has an upper bound of n^2 . You might attempt to bound each term in the summation by the smallest term, but since that term is 1, you would get a lower bound of n for the summation—far off from the upper bound of n^2 .

You can obtain a better lower bound by first splitting the summation. Assume for convenience that n is even, so that

$$\begin{aligned} \sum_{k=1}^n k &= \sum_{k=1}^{n/2} k + \sum_{k=n/2+1}^n k \\ &\geq \sum_{k=1}^{n/2} 0 + \sum_{k=n/2+1}^n \frac{n}{2} \\ &= \left(\frac{n}{2}\right)^2 \\ &= \Omega(n^2), \end{aligned}$$

which is an asymptotically tight bound, since $\sum_{k=1}^n k = O(n^2)$.

For a summation arising from the analysis of an algorithm, you can sometimes split the summation and ignore a constant number of the initial terms. Generally, this technique applies when each term a_k in a summation $\sum_{k=0}^n a_k$ is independent of n . Then for any constant $k_0 > 0$, you can write

$$\begin{aligned} \sum_{k=0}^n a_k &= \sum_{k=0}^{k_0-1} a_k + \sum_{k=k_0}^n a_k \\ &= \Theta(1) + \sum_{k=k_0}^n a_k, \end{aligned}$$

since the initial terms of the summation are all constant and there are a constant number of them. You can then use other methods to bound $\sum_{k=k_0}^n a_k$. This technique applies to infinite summations as well. For example, let's find an asymptotic upper bound on $\sum_{k=0}^{\infty} k^2/2^k$. The ratio of consecutive terms is